Live and Let Sigma Handout

Sigma Rules Workshop

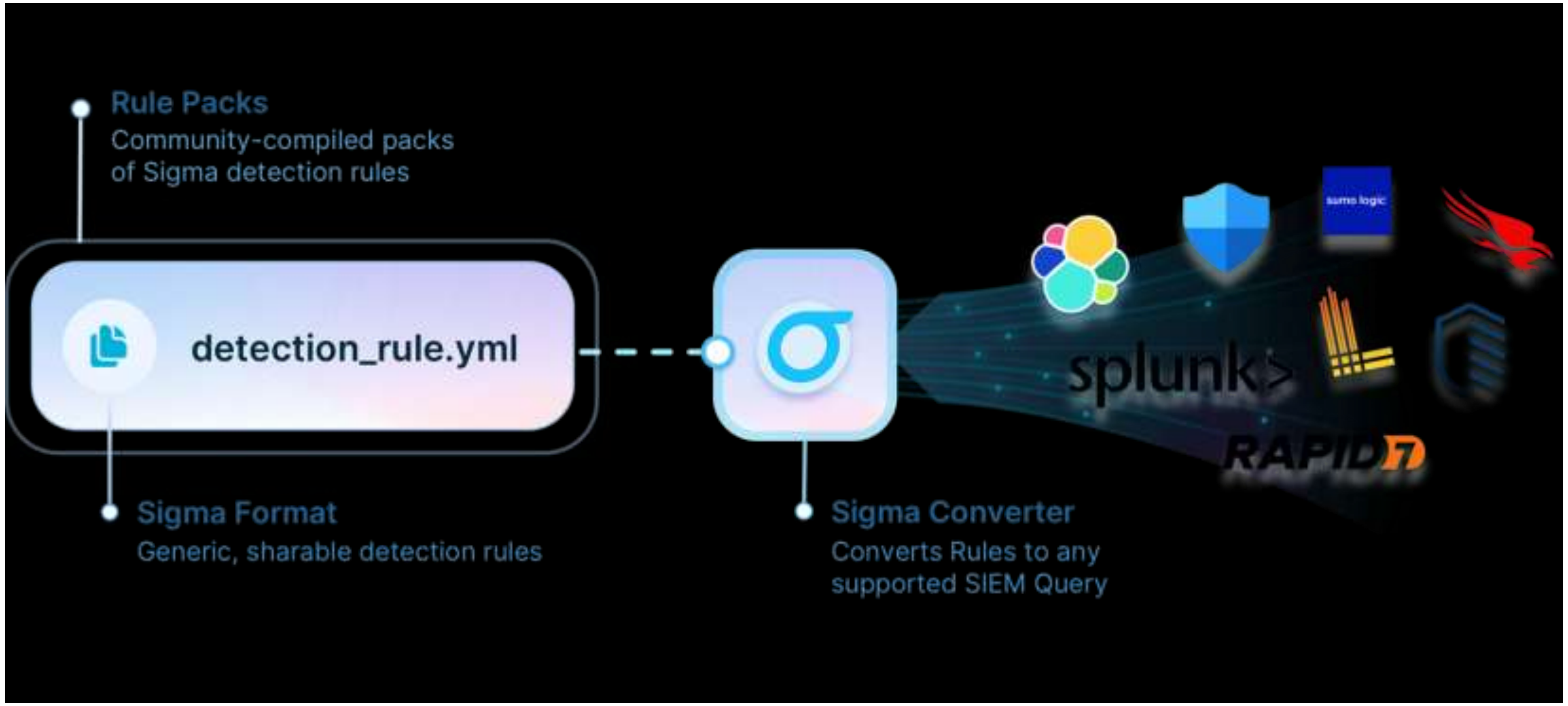
1. Introduction to Sigma

"Sigma is for log files

what Snort is for network traffic

and YARA is for files."

- Quote from sigmahq.io



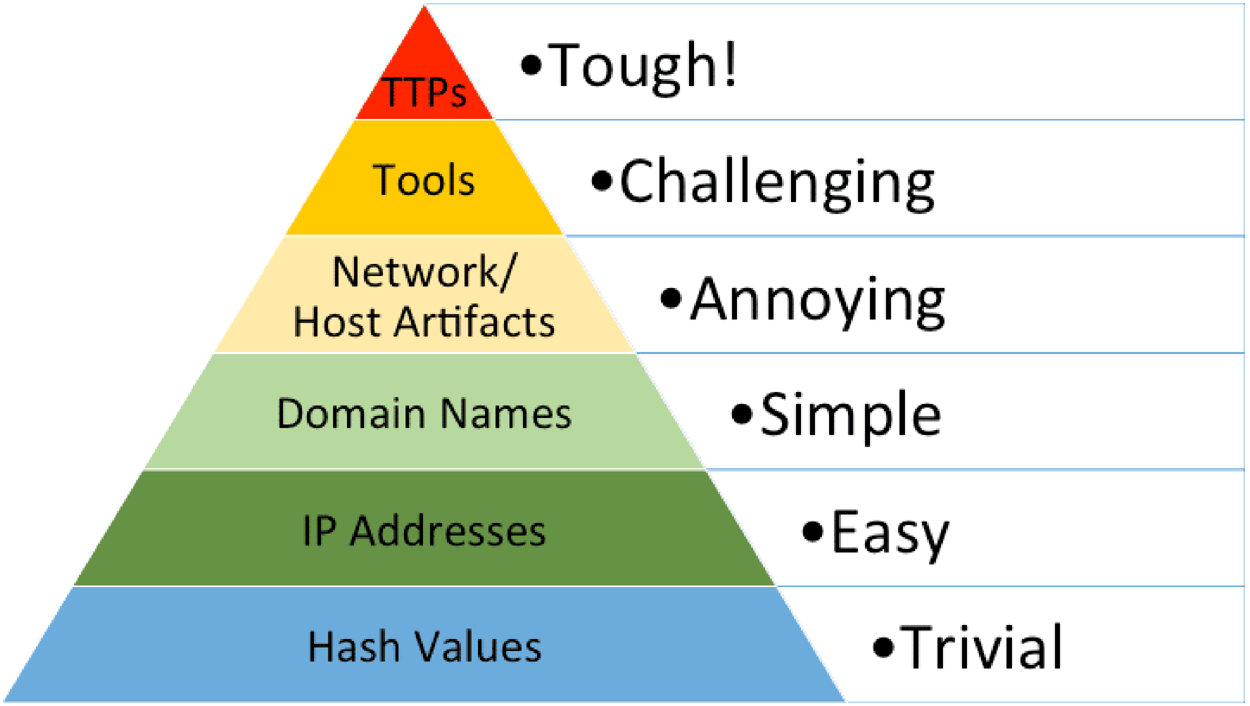
3. Sigma Convertor – Quick demo

1. pip install sigma-cli (ALREADY done in your VM)
2. Open cmd.exe and go to `C:\custom_sigma_rules`
3. Test sigma-cli by using command: `sigma version` (output: 2.0.X)
4. Check your directory has sample sigma rule with name interactive_at_job.yml
5. `sigma convert --target splunk --pipeline splunk_windows C:\custom_sigma_rules\interactive_at_job.yml`
6. You should output something like:

```
Image="*\\at.exe" CommandLine="*interactive*"
```

4. Detection Engineering Workflow:

1. Requirements Gathering and Threat Research
2. Rule Creation
3. Validation and Conversion
4. Testing and Tuning
5. Deployment and Monitoring



4.1 Detection Engineering Workflow:

1. Requirements Gathering and Threat Research
   1. Intel Analysis : Review Threat Reports, malware analysis, or incident data to identify specific behaviour or artifacts
   2. Tactics Mapping : Map activity to framework like MITRE ATT&CK
   3. Log Source Identification : Determine which logs capture necessary data
2. Rule Creation
3. Validation and Conversion
4. Testing and Tuning
5. Deployment and Monitoring

4.2 Detection Engineering Workflow:

1. Requirements Gathering and Threat Research
2. Rule Creation
    1. Metadata
    2. Logsource
    3. Detection Logic
3. Validation and Conversion
4. Testing and Tuning
5. Deployment and Monitoring

4.3 Detection Engineering Workflow:

1. Requirements Gathering and Threat Research
2. Rule Creation
3. Validation and Conversion
   1. Syntax Check
   2. Rule conversion
   3. Pipeline Application
4. Testing and Tuning
5. Deployment and Monitoring

4.4 Detection Engineering Workflow:

1. Requirements Gathering and Threat Research
2. Rule Creation
3. Validation and Conversion
4. Testing and Tuning
   1. Simulated Testing : Run converted query in historical or lab environment
   2. Performance Analysis: Check if query is too resource-intensive for SIEM
   3. Tuning : Adjust logic to reduce false positives
5. Deployment and Monitoring

4.5 Detection Engineering Workflow:

1. Requirements Gathering and Threat Research
2. Rule Creation
3. Validation and Conversion
4. Testing and Tuning
5. Deployment and Monitoring
   1. Version Control : Git Repository
   2. Lifecycle Management : Regular reviews to update or deprecate

Aurora EDR

1. What is Aurora ?
2. Lightweight agent that applies Sigma rules on log data in real-time on endpoints
3. Uses ETW (Event Tracing for Windows)
4. Managed locally via config files or via ASGARD Management Center
5. Consider it as custom Sigma-based EDR
6. Extends the Sigma standard with ‘response’ actions
    1. Kill, KillParent, Suspend, Dump
    2. Custom actions

Aurora EDR: **specific Response Actions**

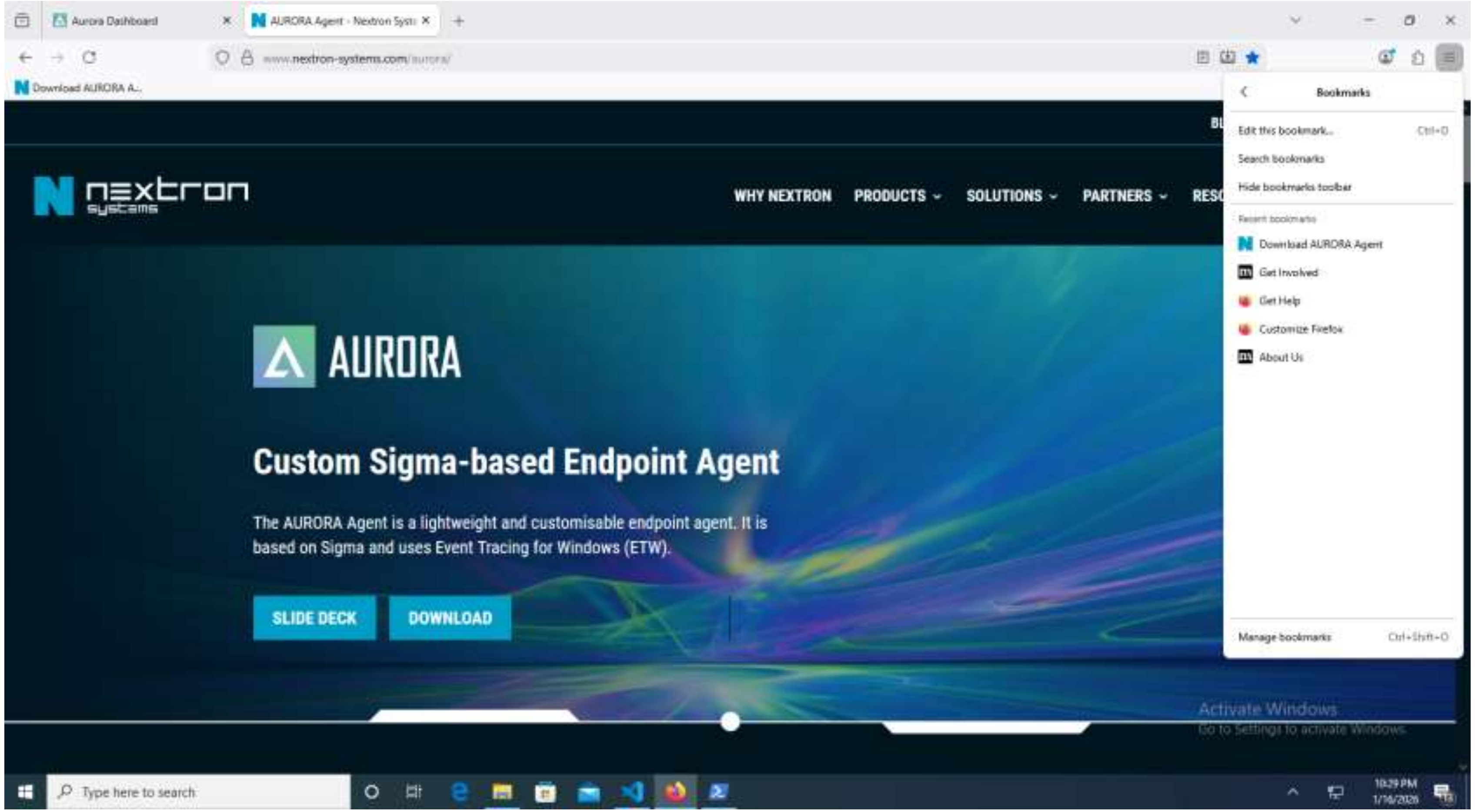
1. In the Sigma rule we can specify response action for Aurora EDR
2. Response action can be:
   1. Predefined
      1. Kill a process or parent process
      2. Suspend a process
      3. Dump Process memory
   2. Custom
      1. A custom command line that can use environment variables and event values
      2. e.g. copy %image% %%ProgramData%%\ProcessId%.bin
3. This can help containment of threat quickly

Installation of Aurora Agent

1. In your Windows 10 VM open C:\ Folder
2. You should see a folder **aurora-agent-lite-win-pack**
3. **Only If you don't the folder follow rest of steps 4,5**
4. You should find zip file with name **aurora-agent-lite-win-pack.zip**
   i. If you do not find above file do get the USB drive and copy this file in your VM
5. Unzip the **aurora-agent-lite-win-pack.zip** file by right click and selecting Extract All menu.
6. Check if c:\ have 4 more folders **attack, custom_sigma_rules, sigma_test_repo, web**
7. In Explorer go inside the folder **C:\aurora-agent-lite-win-pack**

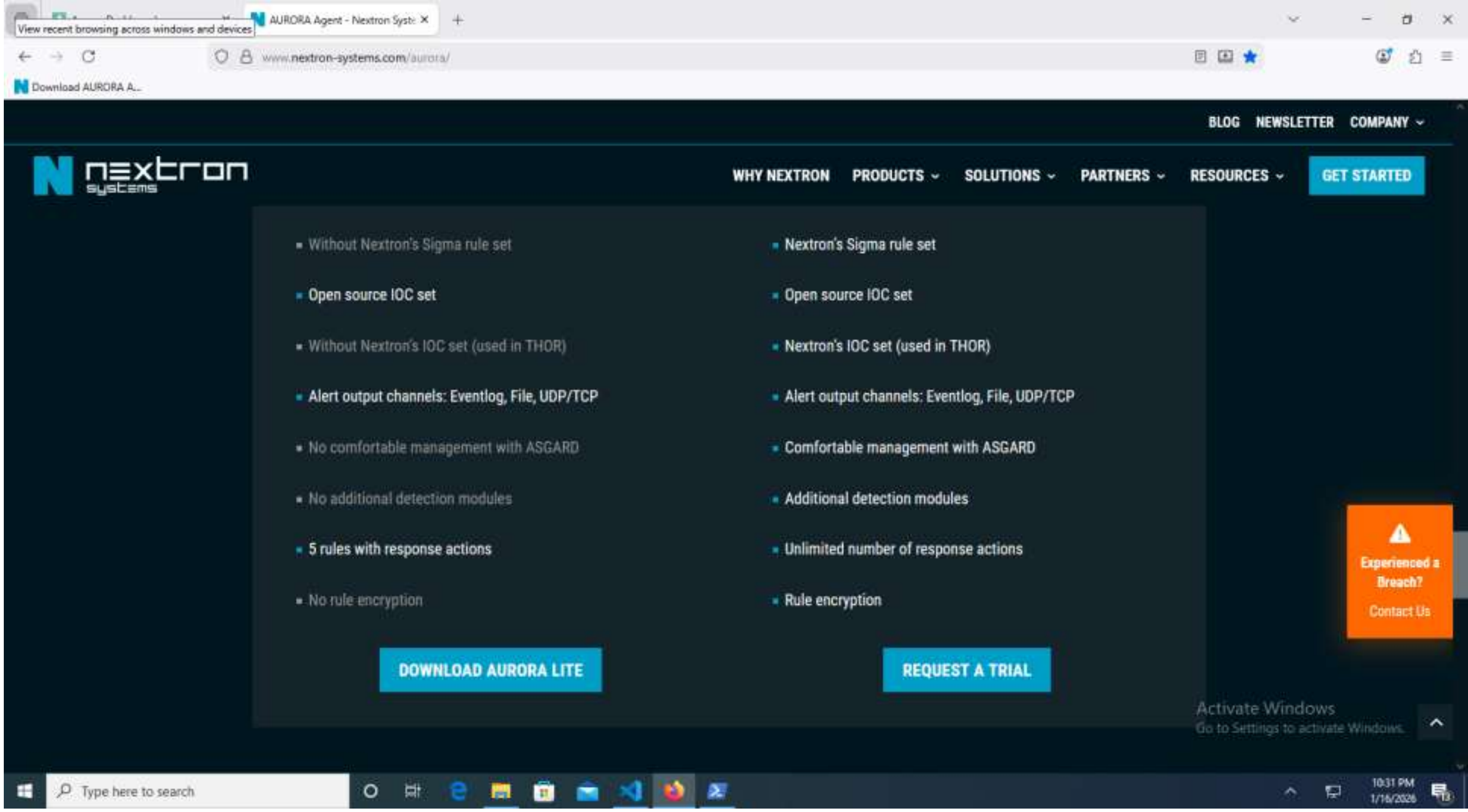
Installation of Aurora Agent

- Open following link: https://www.nextron-systems.com/aurora/
- There should be bookmark "Download AURORA Agent" in your browser for this not open by typing URL
- Click Download



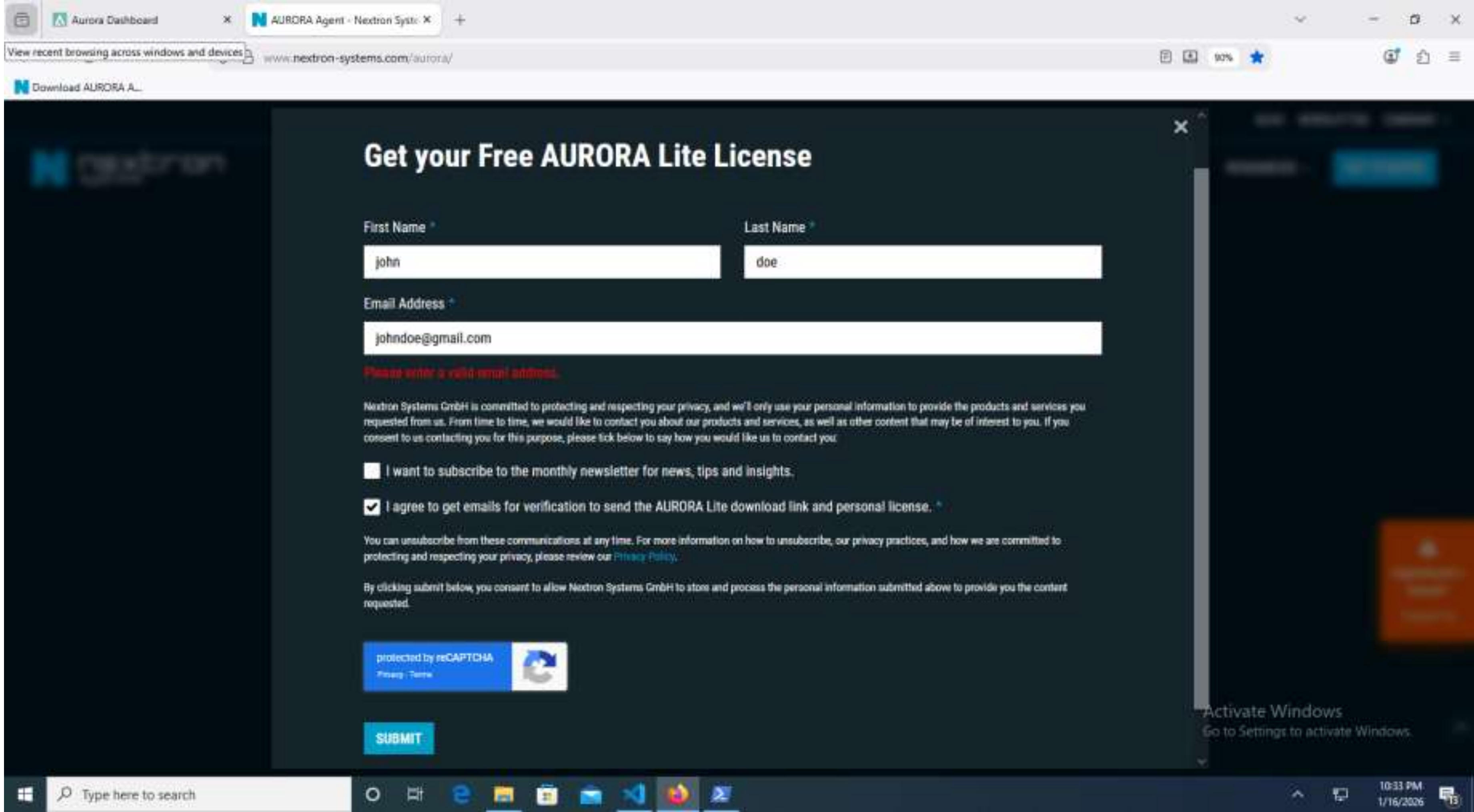
Installation of Aurora Agent

- Click on button DOWNLOAD AURORA LITE



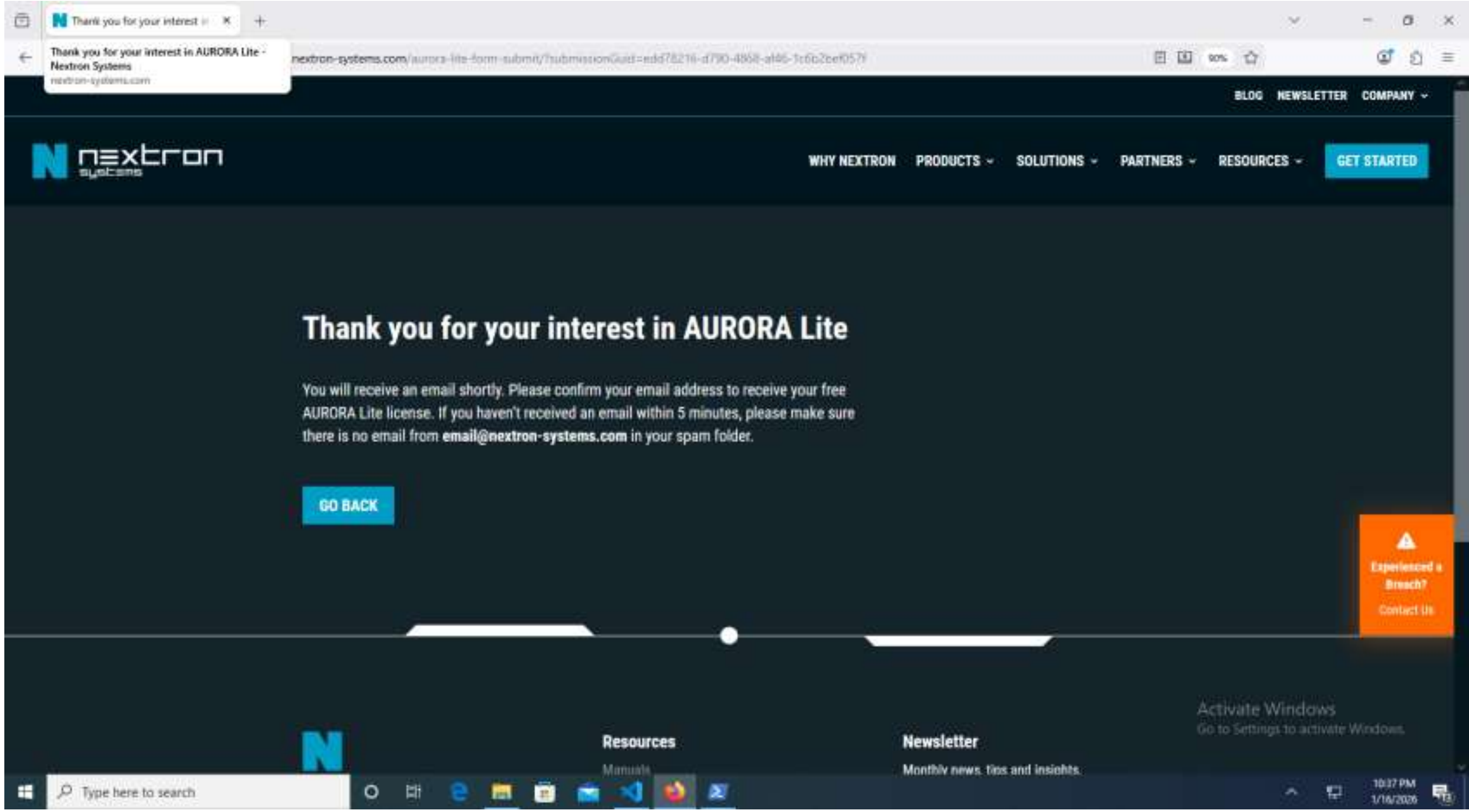
Installation of Aurora Agent

- Fill in any name but use email where you can get email
- You can use disposable email also if you want



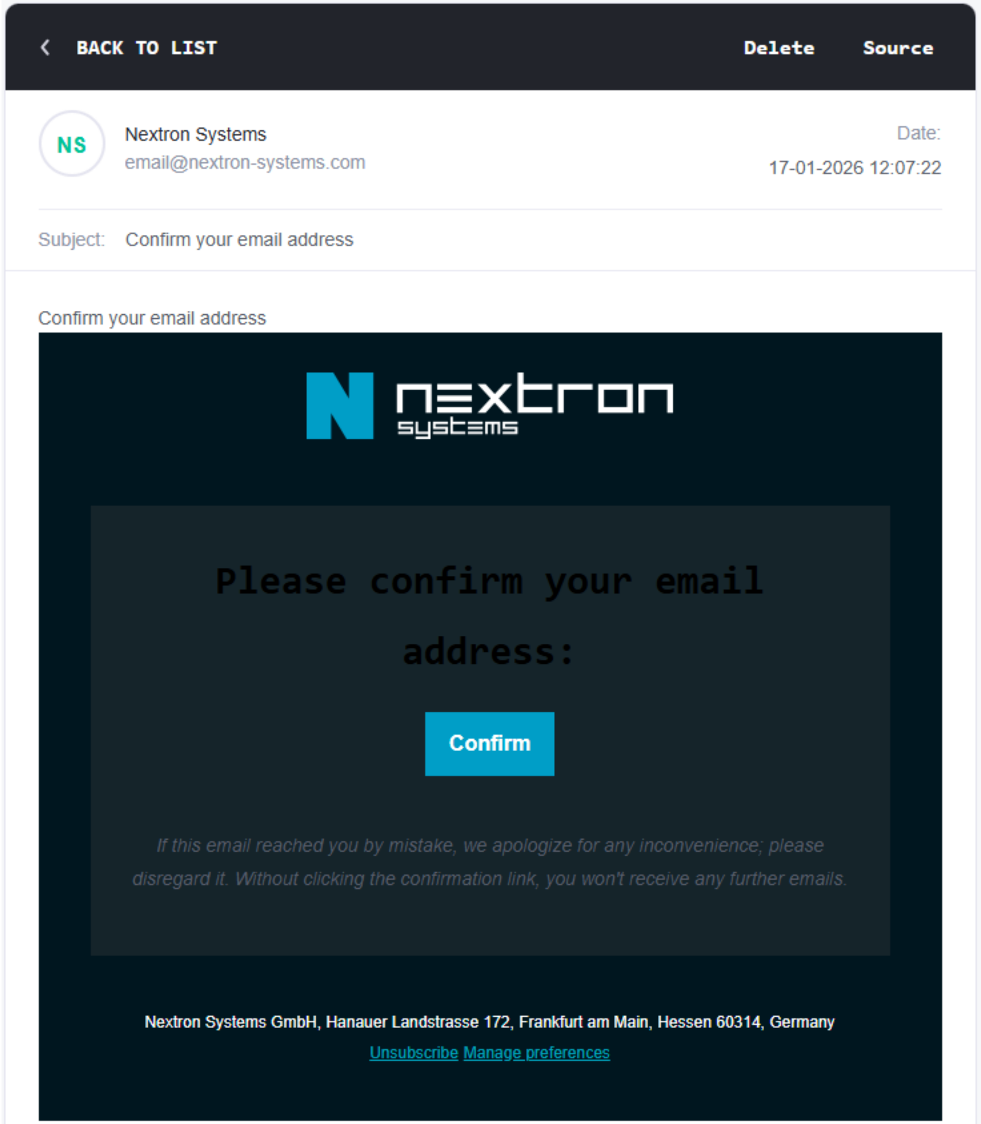
Installation of Aurora Agent

- You Should see something like this Thank You message:



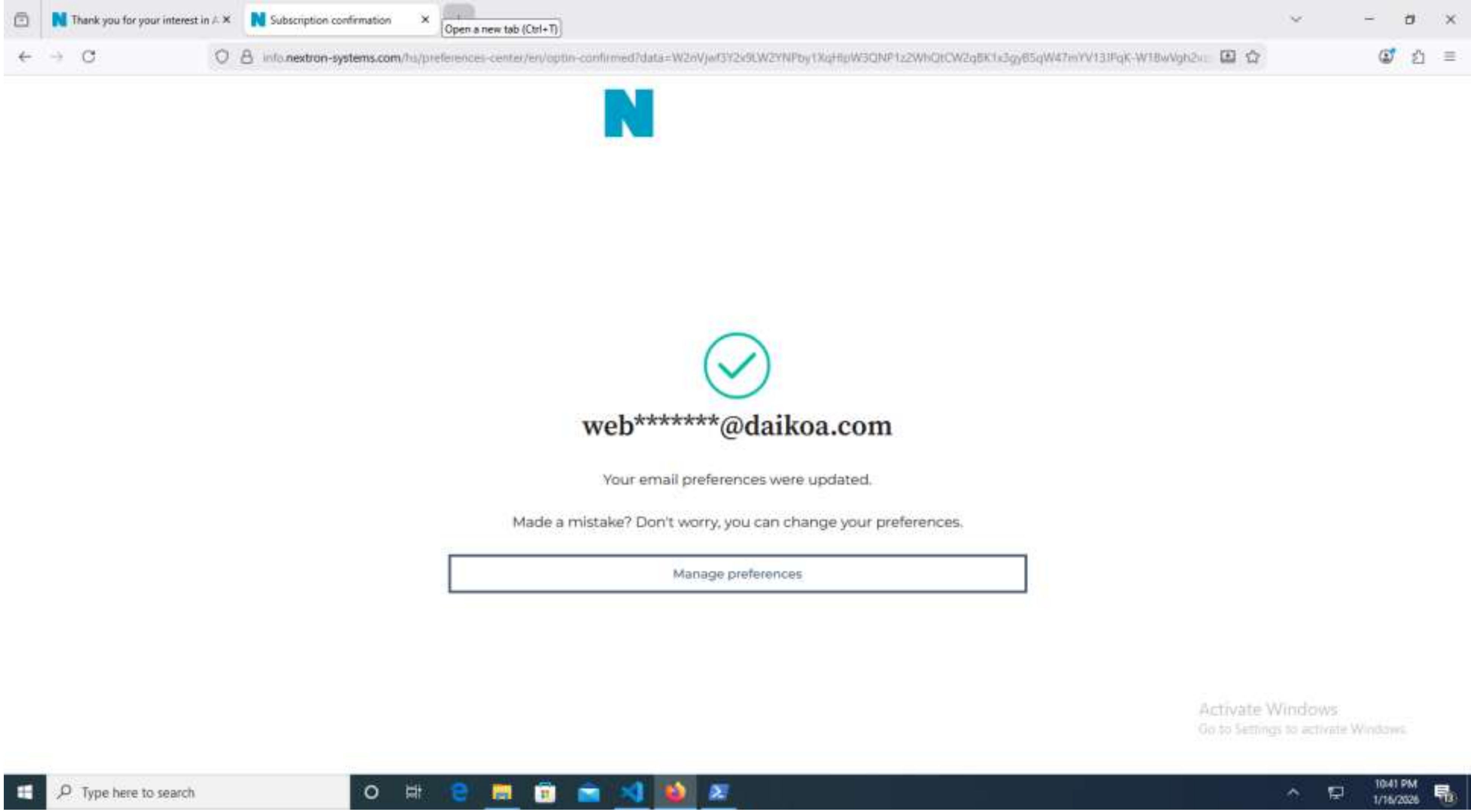
Installation of Aurora Agent

- You will receive a email like this click on “Confirm” button



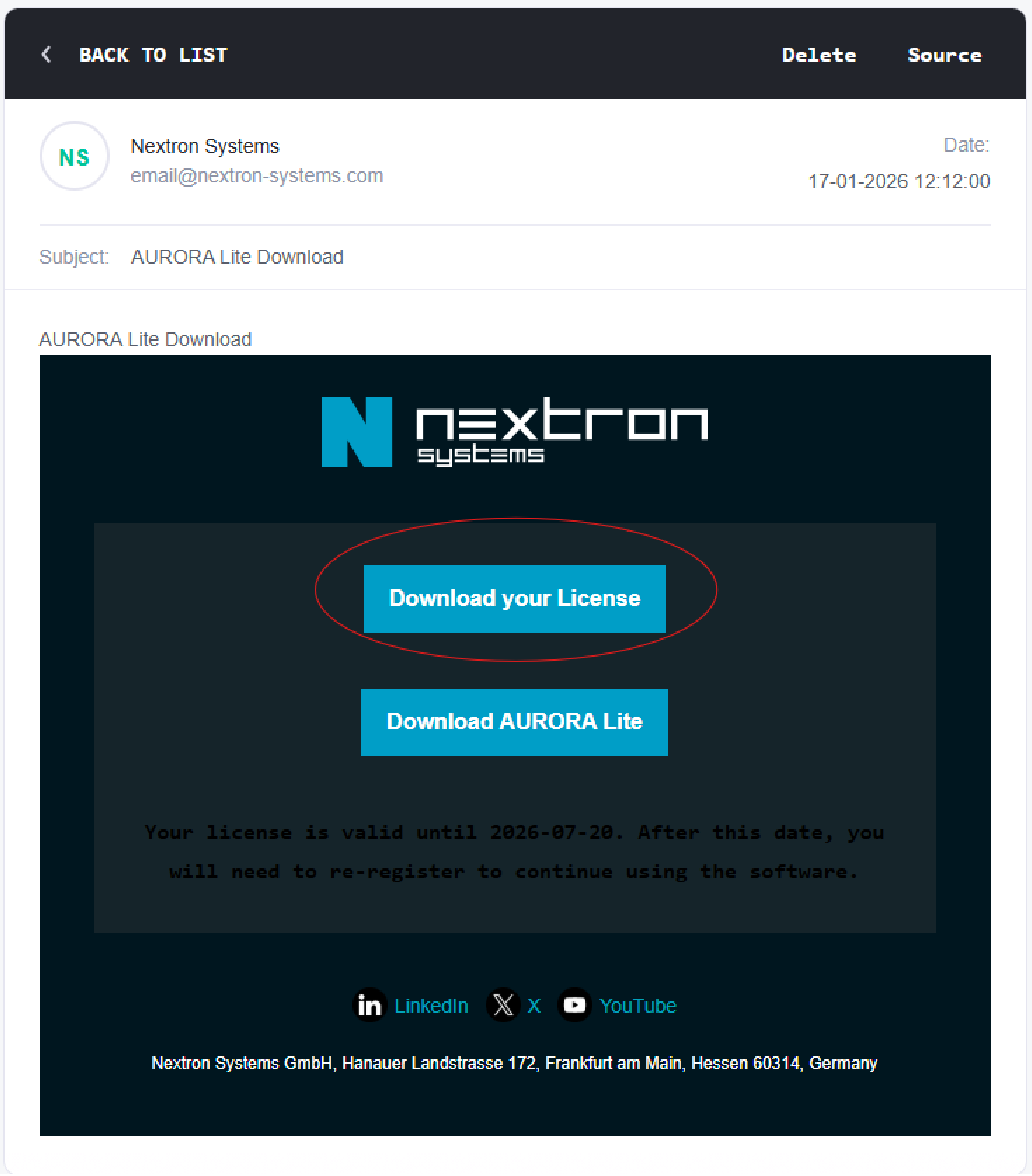
Installation of Aurora Agent

- You will receive a email like this click on "Confirm" button



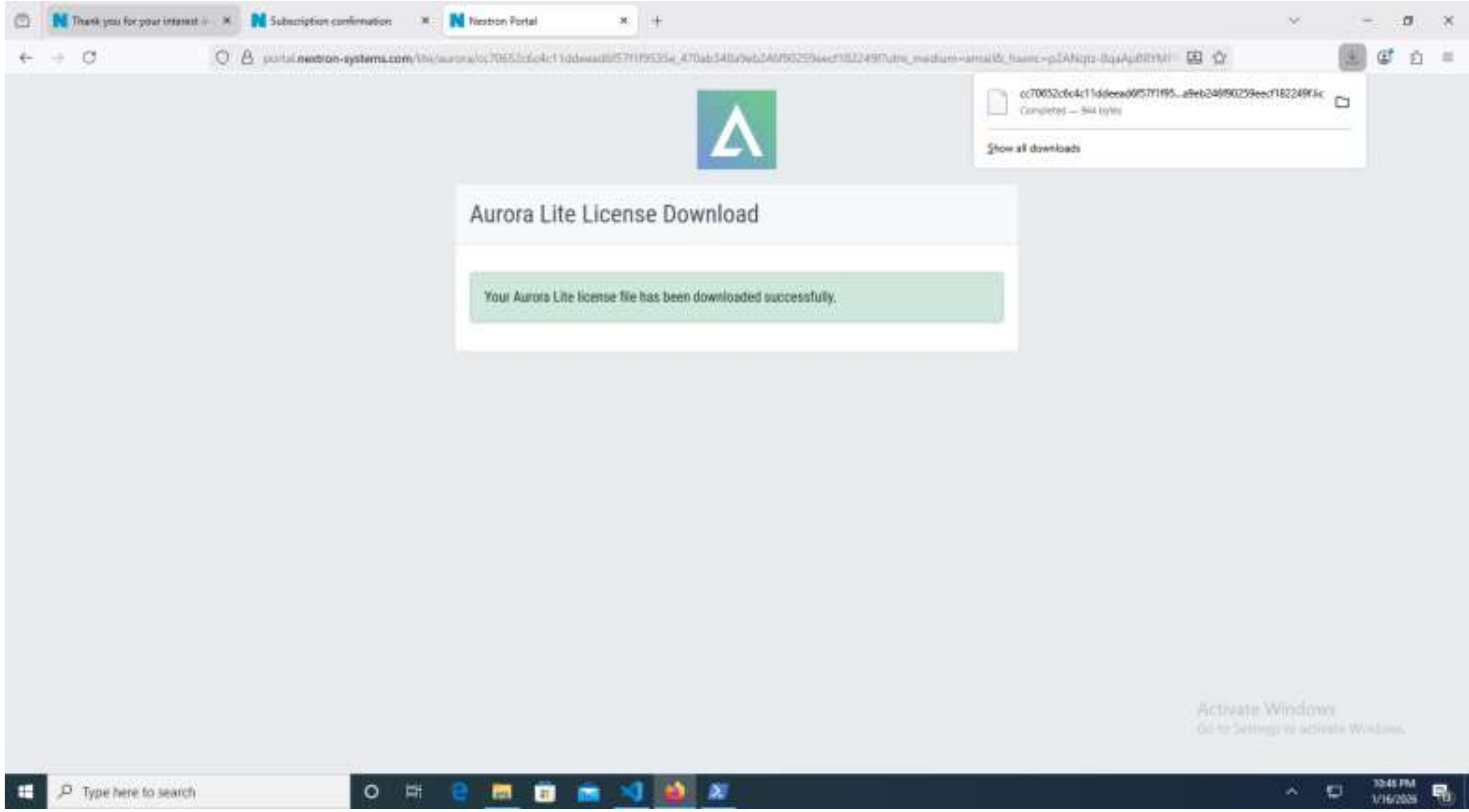
Installation of Aurora Agent

- You will receive a email with subject: AURORA Lite Download
- Click on Download your Licence
- If you are outside your VM (on host) for internet access then copy this .lic file to your Windows 10 VM



Installation of Aurora Agent

- Copy the downloaded .lic file in to the extracted AURORA folder in Downloads directory
- C:\aurora-agent-lite-win-pack\



Installation of Aurora Agent

- In Windows Explorer Click on File and Open Window PowerShell -> Open Window PowerShell as Administrator
- Using “dir” command verify .lic file is in same extracted folder

Administrator: Windows PowerShell

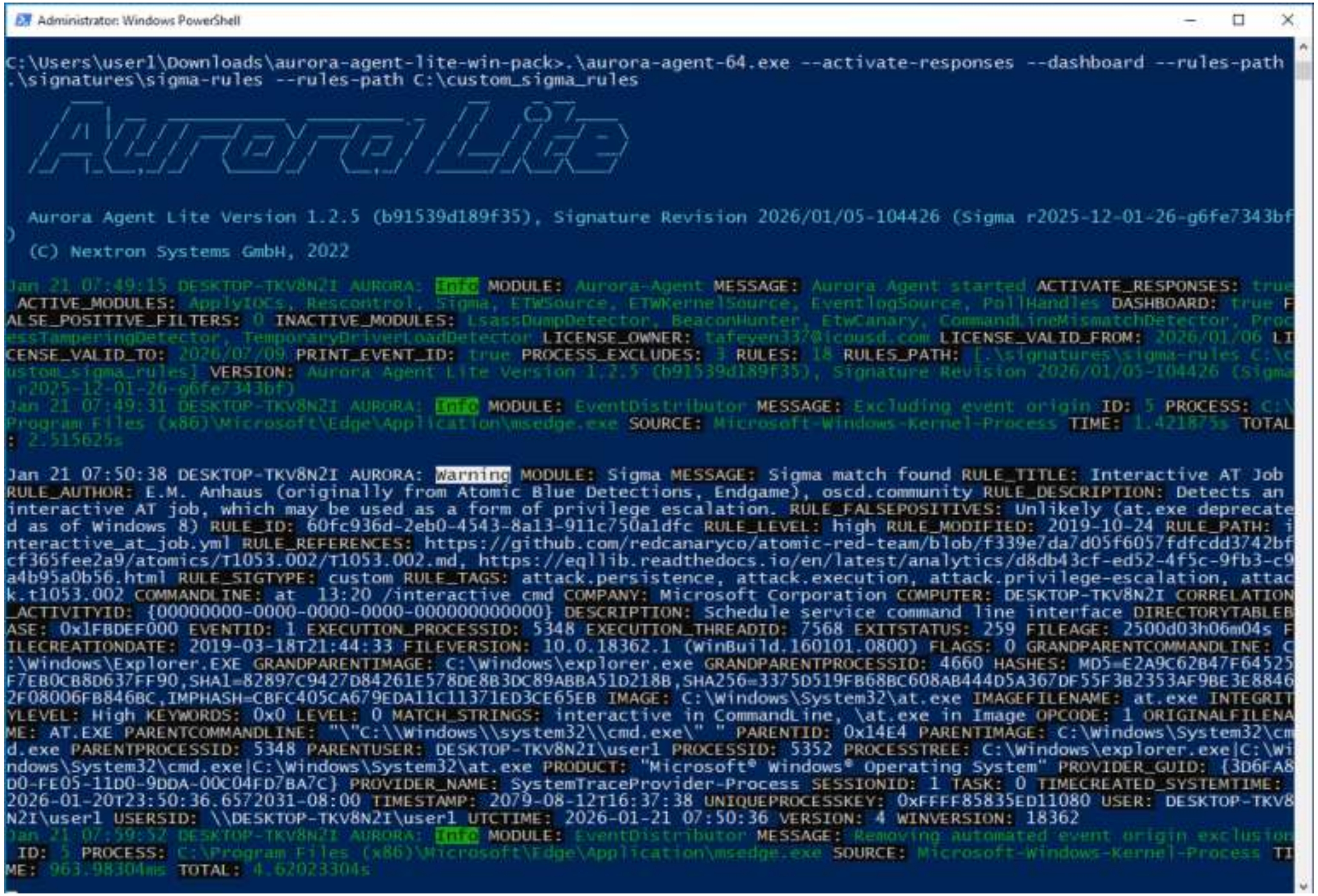
```
PS C:\Users\user1\Downloads\aurora-agent-lite-win-pack> dir


    Directory: C:\Users\user1\Downloads\aurora-agent-lite-win-pack


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----          1/6/2026   7:08 AM                config
d-----          1/6/2026   7:08 AM                custom-signatures
d-----          1/6/2026   7:08 AM                docs
d-----          1/6/2026   7:08 AM                ioc-configs
d-----          1/6/2026   7:08 AM                log-sources
d-----          1/6/2026   7:08 AM                response-sets
d-----          1/6/2026   7:07 AM                signatures
d-----          1/6/2026   7:08 AM                tools
-a----          1/6/2026   7:03 AM            944 89f358182f97a8b40fb8cbe3f96bde63_cd9568a5fbfbd79cde94a8435118a480
                                                  .lic
------         9/23/2025   8:01 PM            347 agent-config-intense.yml
------         9/23/2025   8:01 PM            463 agent-config-minimal.yml
------         9/23/2025   8:01 PM            369 agent-config-reduced.yml
------         9/23/2025   8:01 PM            320 agent-config-standard.yml
------         9/23/2025   8:01 PM       16609080 aurora-agent-64.exe
------         9/23/2025   8:01 PM       13976376 aurora-agent-util.exe
------         9/23/2025   8:01 PM       15715640 aurora-agent.exe


PS C:\Users\user1\Downloads\aurora-agent-lite-win-pack>
```

Installation of Aurora Agent

- Run Agent by using command:

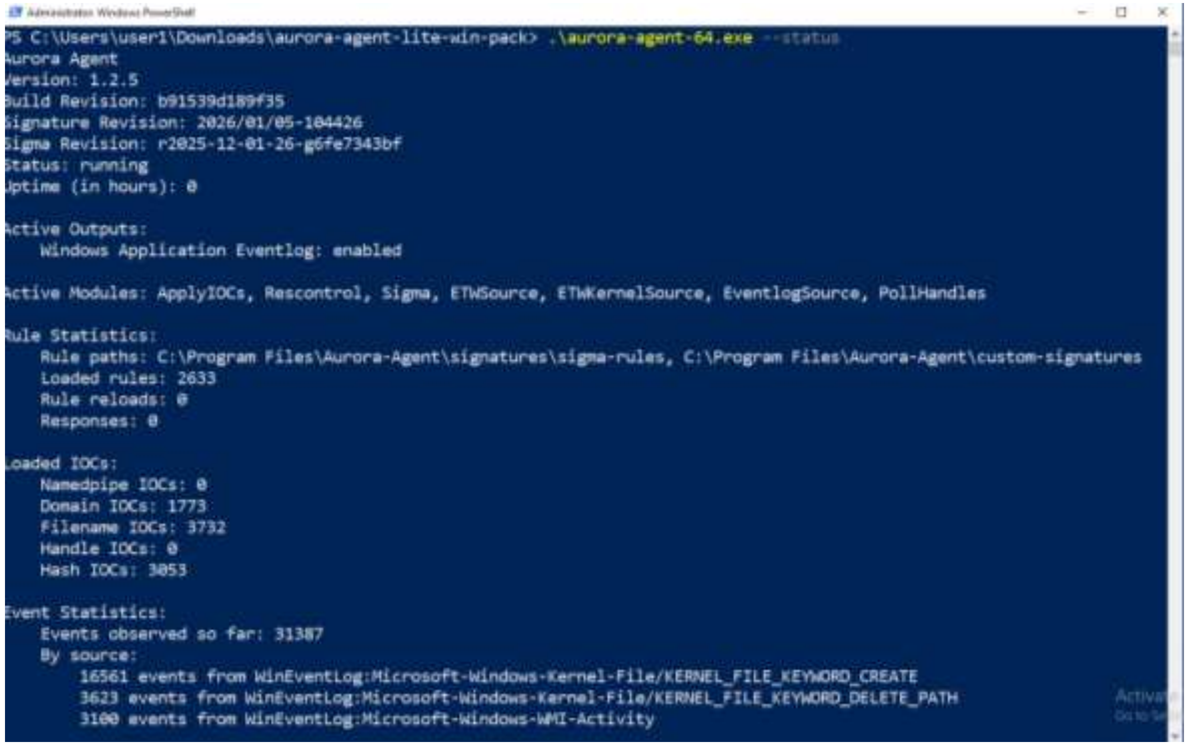
```
.\start_custom_agent.bat
```

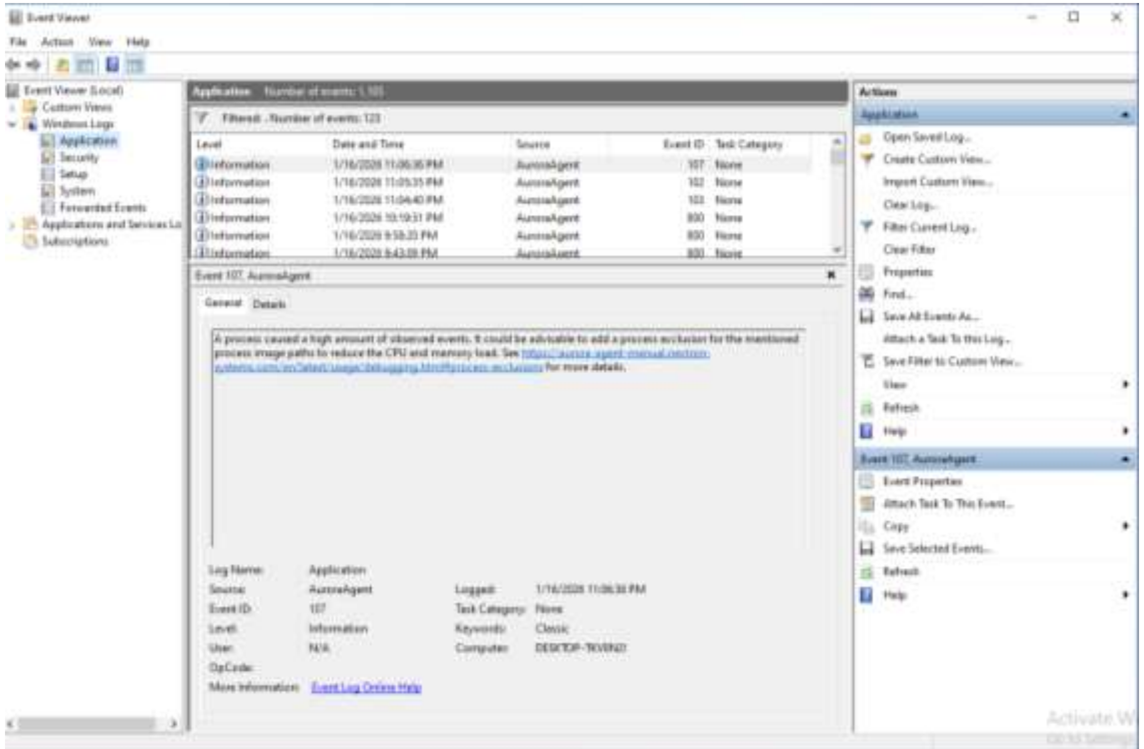
- You should see output like:



Installation of Aurora Agent

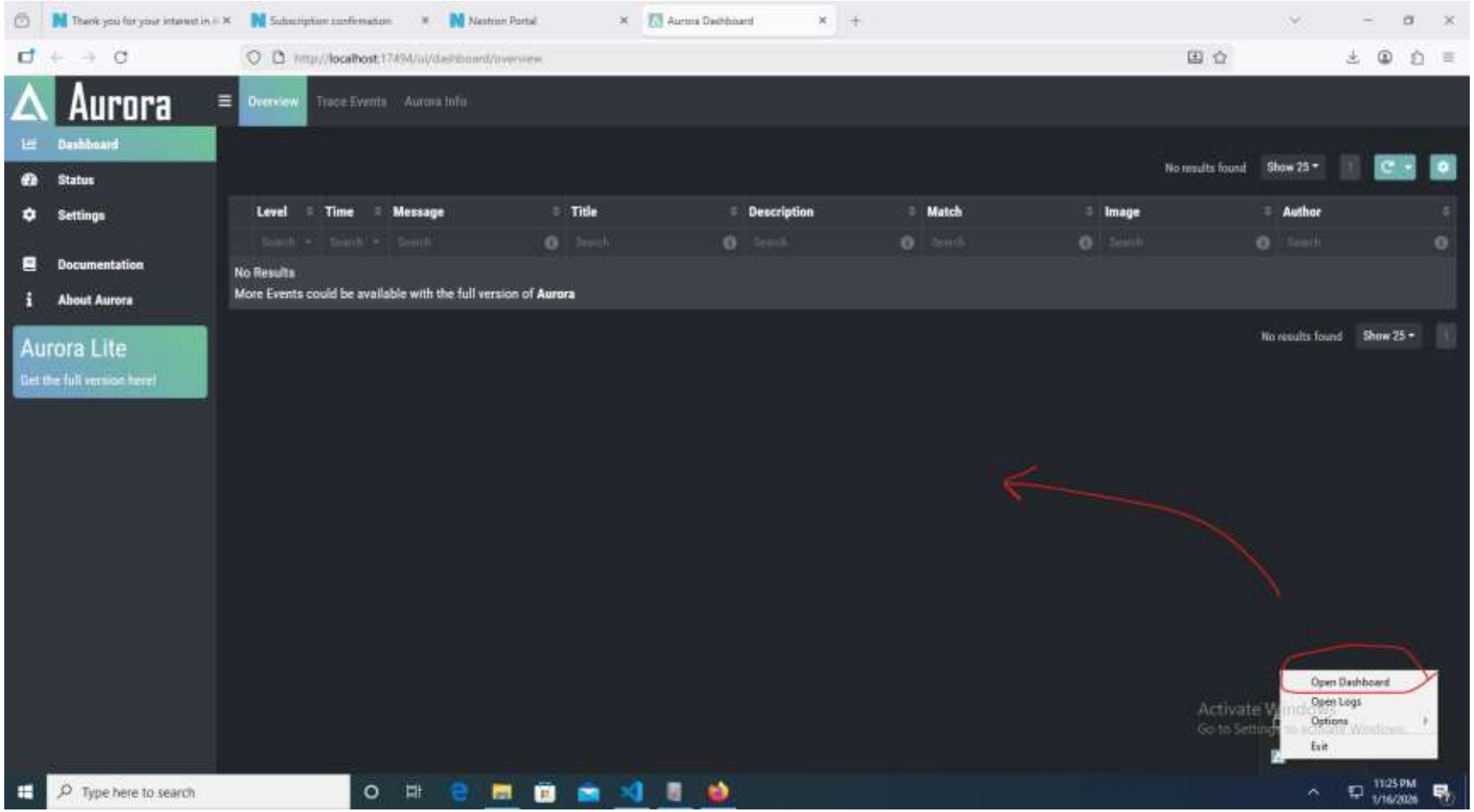
- Open new PowerShell window and Verify installation by using command:
  `.\aurora-agent64.exe --status`
- Verify new events in the local “Application” event log (Event Viewer)
  - Win+R --type--> eventvwr.msc
  - EventViwer (Local) -> Windows Logs -> Application





Installation of Aurora Agent

- Right click on aurora icon near bottom right corner
- Click on menu

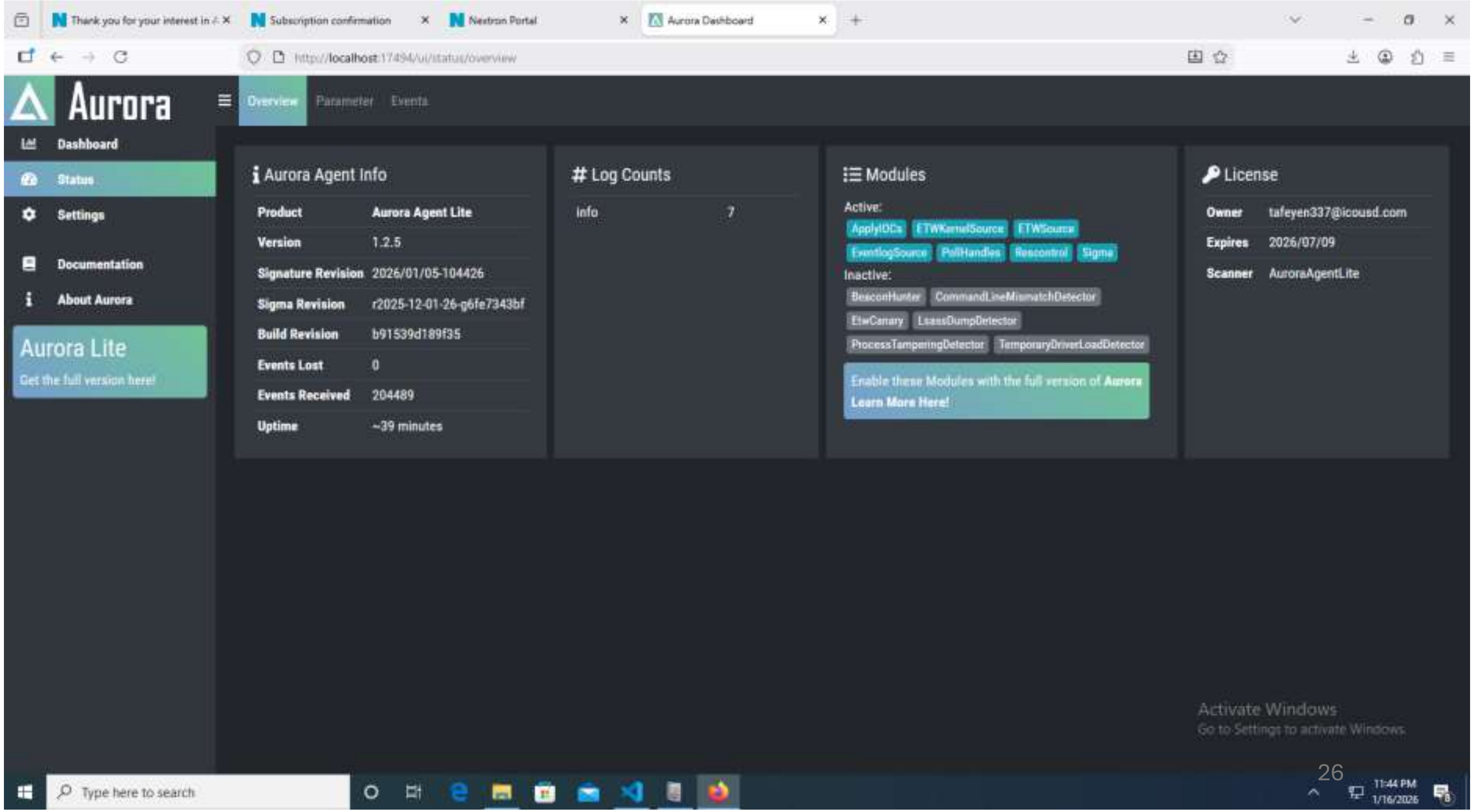


Installation of Aurora Agent

- If default click on dashboard does not open the browser do following:
  - Open Edge/ FireFox browser
  - In browser open URL **http://localhost:17494/ui/dashboard/overview**

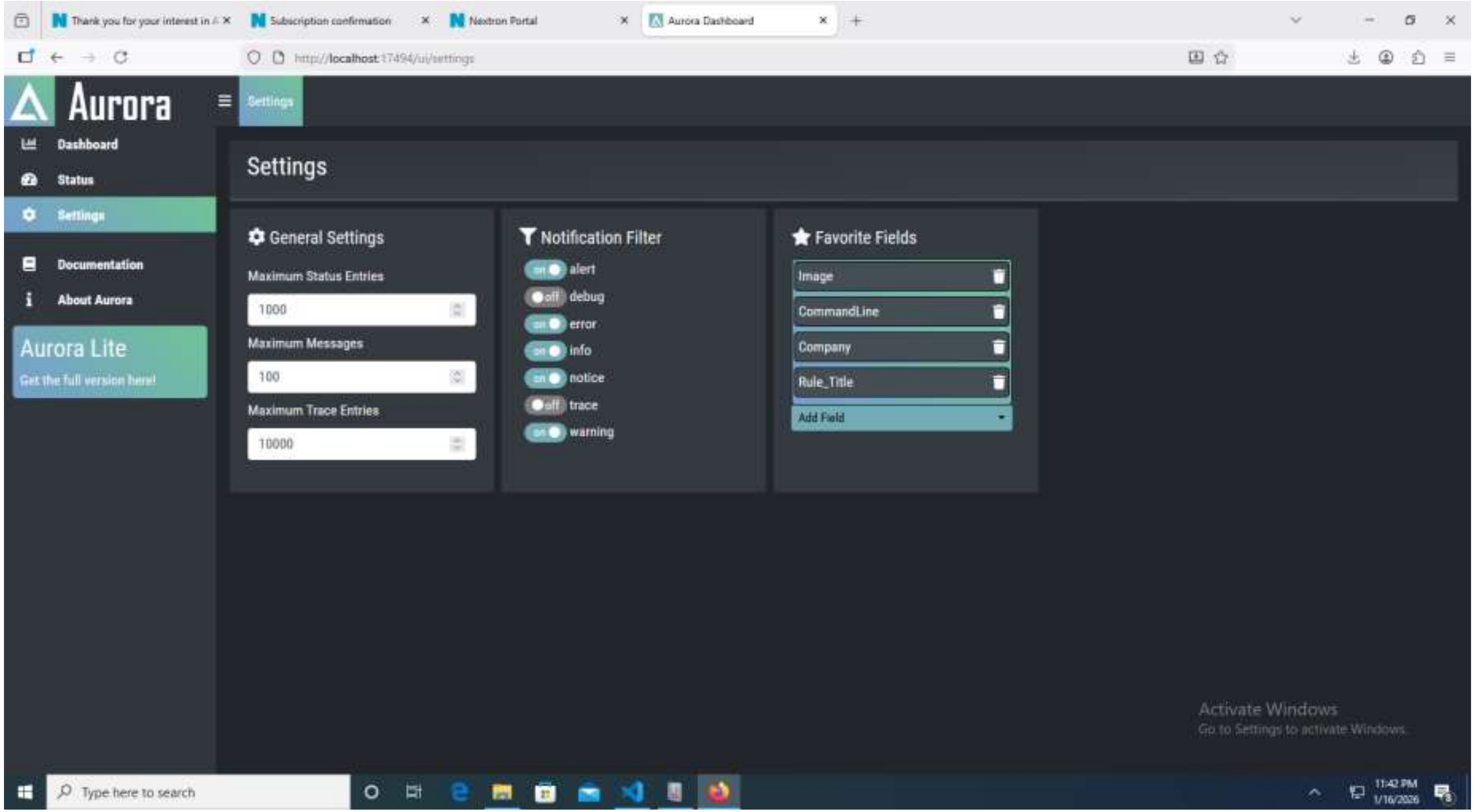
Installation of Aurora Agent

Status Tab:



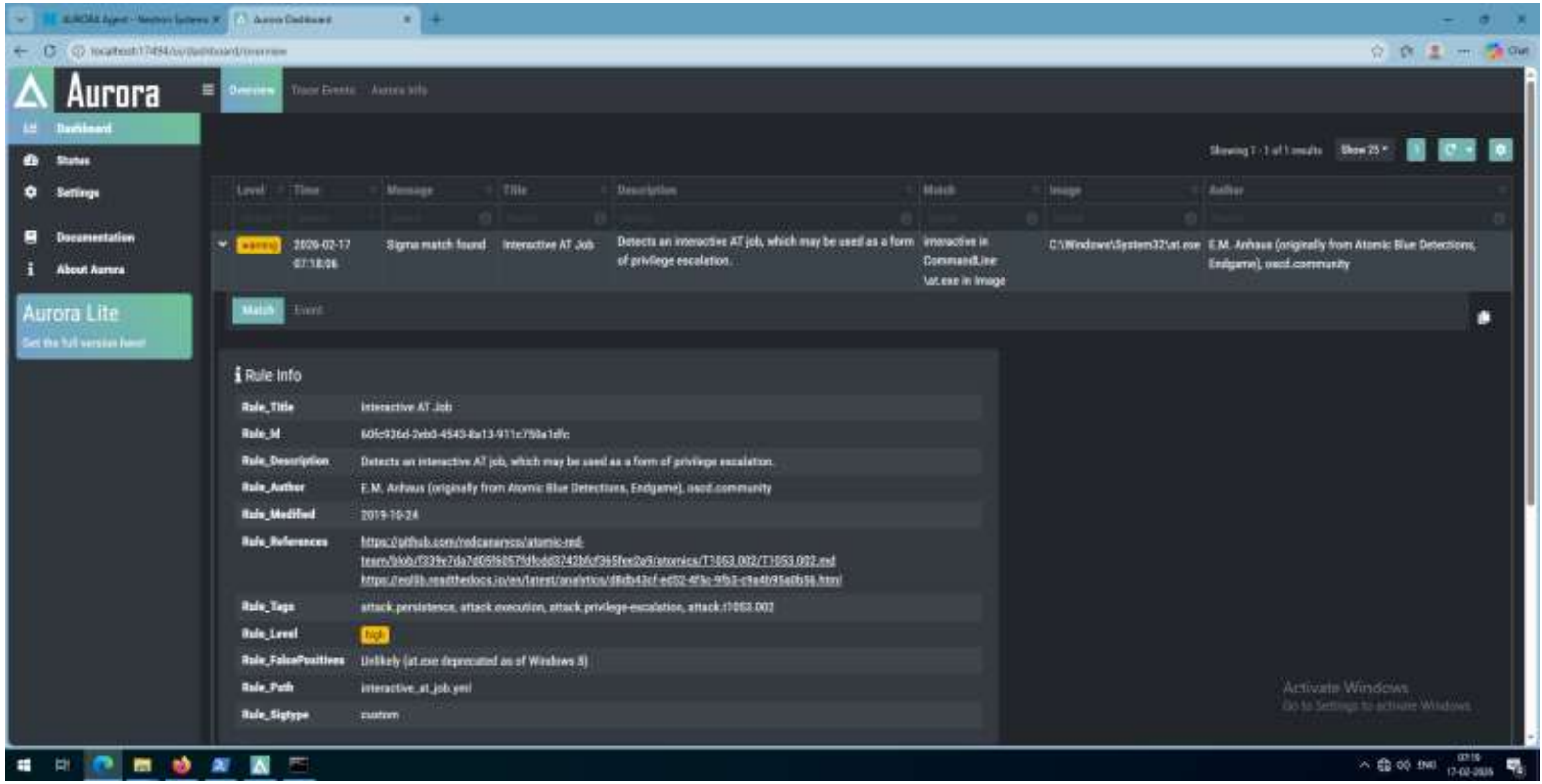
Installation of Aurora Agent

Settings Tab:



Installation of Aurora Agent – Test Installation

1. Just execute command:
   1. at 14:30 /interactive cmd.exe
2. Open Dashboard section and OverView tab
3. You should see one alert with title Interactive AT job



First Sigma Rule

1. We will start with a widely used persistence technique
2. T1547.001 : Boot or Logon AutoStart Execution: Registry Run Keys
   1. Adversaries may achieve persistence by adding a program to a startup folder or referencing it with a Registry run key. Adding an entry to the "run keys" in the Registry or startup folder will cause the program referenced to be executed when a user logs in
3. Execute following commands
   1. `Windows + R -> cmd.exe`
   2. `cd c:\attack`
   3. `attack01.bat`
4. BAT output will also show contents of HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\Run

First Sigma Rule

- Open registry editor by using following commands
  - `Windows + R`
  - `regedit (press enter)`
- Open Path Computer\HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run
- Check the registry keys – which one seem different ?
- Lets create a basic Sigma rule to capture creation of this registry key

First Sigma Rule

1. Open VSCode / Notepad++ in your Windows 10 VM
2. File -> New File ->
3. File -> Save -> C:\custom_sigma_rules\my_sigma_rule.yml
4. Run following python script to get unique UUID4 ID:
    1. `python c:\sigma_test_repo\get_uuid4.py`
5. Enter Rule values as shown in the next slide

First Sigma Rule

- ID: Use UUID4 generated in previous step (unique for each)
- You can copy from output of attack01.bat the Registry key path
- Keep in mind the same space for same level elements

```yaml
title: My First Sigma Rule Capture New Registry Entry To Autostart
author: xyz
description: "This will detect reg.exe execution with add parameter"
id: 52e815fe-7349-453d-86fe-16fd9f96f5b5
level: high
status: experimental
date: 2026-01-27
logsource:
  category: process_creation
  product: windows
detection:
  selection:
    Image|endswith: '\reg.exe'
    CommandLine|contains|all:
      - 'reg'
      - ' add '
    CommandLine|contains:
      - 'Software\Microsoft\Windows\CurrentVersion\Run'
  condition: selection
```

First Sigma Rule

1. Run following python script to validate our new rule:
   1. `python C:\sigma_test_repo\validate_sigma.py C:\custom_sigma_rules\my_sigma_rule.yml`
2. Ideal condition you should see output like:
   1. Rule 'C:\custom_sigma_rules\my_sigma_rule.yml' is valid!
3. If you want to check how errors look change the case of letter ‘I’ in Image|endswith
4. Run the validate_sigma.py script with same command above you will see something like:
   1. `[SigmahqInvalidFieldnameIssue] issue=SigmahqInvalidFieldnameIssue severity=high description="A field name do not exist" rules=[52e815fe-7349-453d-86fe-16fd9f96f5b5] field=aaImage Details: A field name do not exist`

7.1 Rule Testing

1. Newly created rule .yml file is in the C:\custom_sigma_rules directory
2. Now we will stop the Aurora Agent
   1. Press Ctrl+C
   2. Press Y to stop batch job
3. If closed – Re Open Command / PowerShell
4. Restart Aurora agent by executing command:
   1. `.\start_custom_agent.bat`
5. Again Execute attack by using following commands
   1. `Windows + R -> cmd.exe`
   2. `cd c:\attack`
   3. `attack01.bat`

7.1 Rule Testing

- Now open the Aurora Agent management console
- Open the Dashboard section on left menu
- Click on Overview Tab on top to see generated events
- You should see event with same title in rule file